M-26-14 Elastic reference architectures

Reference architectures for meeting OMB M-26-14 logging and retention requirements using the Elasticsearch Platform on Elastic Cloud Hosted

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Regulatory context: OMB M-26-14

Issued May 22, 2026, [OMB Memorandum M-26-14] — Ensuring Effective and Efficient Agency Logging and Network Visibility to Defend Against Evolving Cyber Threats — establishes the current federal standard for agency logging and log retention. M-26-14 rescinds the prior M-21-31 logging mandate and replaces it with a risk-based, outcome-oriented framework organized around two distinct operational objectives:

- **Continuous Event Monitoring (CEM):** Real-time log ingestion and alerting, focused on detecting anomalous activity as it occurs
- **Threat Hunting, Investigation, Response, and Forensics (THIRF):** Longer-horizon retention supporting forensic reconstruction, incident investigation, and threat hunting across extended time windows

M-26-14 establishes absolute minimum baseline requirements of six months searchable and twelve months retrievable (Appendix B). However, these baselines are reached progressively through a five-level maturity model (Appendix C) that governs the Data Retention element alongside inventory visibility, collection coverage, collection operations, and log management. Retention obligations scale with maturity level — from retrievable-only at six months at Level 1, through a combination of searchable and retrievable windows at Levels 3 and 4 — with the full six-month searchable and twelve-month retrievable requirement representing the Optimal (Level 4) target. Agencies are required to achieve Level 1 within 120 days of the CISA Logging Reference Architecture publication, Level 2 within 180 days, and Level 3 within 320 days.

The tiered storage architecture described in this document is designed to support agencies at any point in that maturity progression — providing the retention depth and storage economics to meet current obligations while scaling to satisfy the Optimal threshold as agencies advance through the model.

Elastic overview

The Elasticsearch Platform is a distributed, real-time analytics engine purpose-built to ingest, store, search, and analyze structured and unstructured data at enterprise scale. At its core, Elasticsearch provides the indexing and query engine that powers the platform, while Kibana delivers interactive visualization and operational dashboards, Elastic Agent standardizes telemetry collection across diverse infrastructure, and a broad library of integrations accelerates onboarding from virtually any data source.

A defining characteristic of the platform is its support for both schema-on-write and schema-on-read workflows, enabling teams to normalize data at ingest time via the Elastic Common Schema (ECS) or defer normalization to query time — a flexibility that significantly reduces the time required to operationalize new data sources and correlate events across domains.

The platform's tiered storage architecture — spanning hot, cold, and frozen data tiers — allows organizations to manage data economics alongside data volume. Frequently queried data is served from high-performance local storage, while older indices transition automatically to lower-cost tiers and ultimately to object storage via searchable snapshots, enabling long-term retention without proportional cost growth. Native machine learning capabilities, including anomaly detection, inference pipelines, and data frame analytics, are available as first-class platform features and can be isolated to dedicated node capacity to prevent contention with search and ingest workloads.

Elastic Cloud Hosted (ECH) extends these capabilities into a fully managed deployment model available across major cloud providers, with support for FedRAMP High and Moderate authorization boundaries — making the platform suitable for federal and regulated enterprise environments that require both operational scale and compliance rigor. For organizations with on-premises requirements or existing infrastructure commitments, Elastic's self-managed deployment option provides the same platform capabilities without a cloud dependency.

Reference architecture

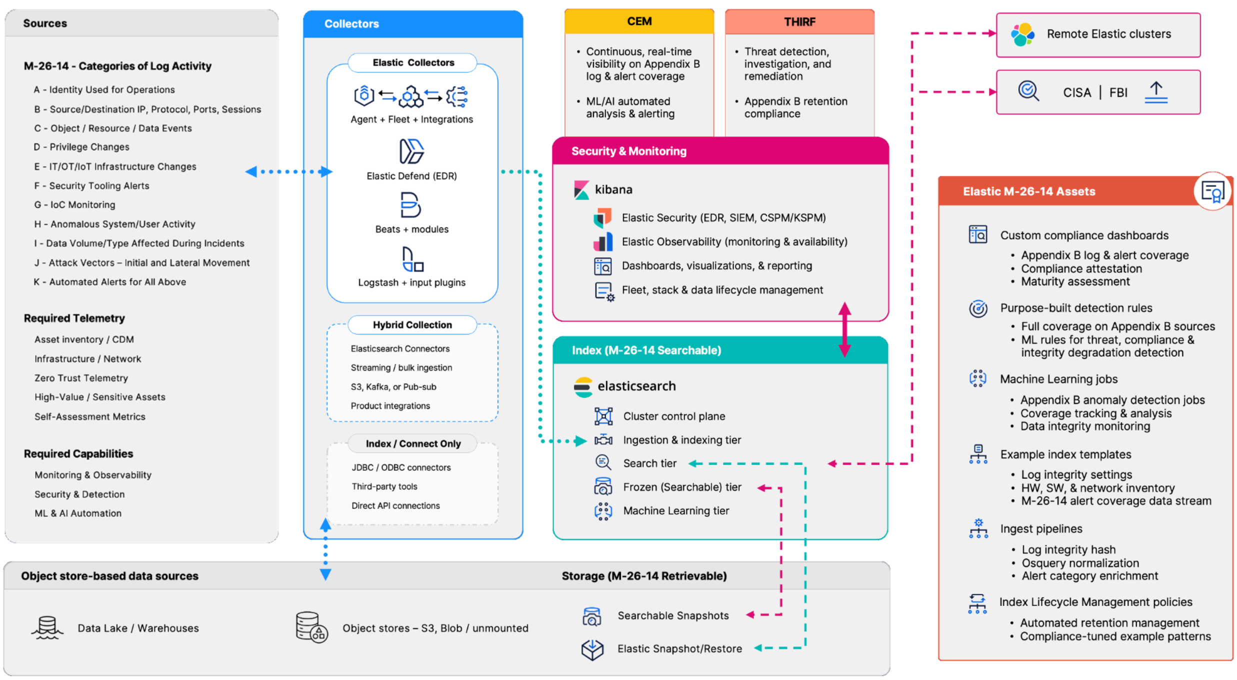

Figure 1: *Elastic provides full coverage for M-26-14 collection and compliance*

Flexible integration options

The reference architecture in Figure 1 depicts the Elasticsearch Platform providing the foundational search, retention, and retrieval functions across all M-26-14 categories of log data streams and required capabilities. The diagram includes the core Elasticsearch Platform components and a summary of the assets provided with the Elastic M-26-14 Compliance Pack; what's not shown are the multiple flexible data collection, integration, deployment, and system monitoring options available within that architecture.

Elastic deployment options

The Elasticsearch Platform is extremely flexible in terms of supported infrastructure and can deploy components to various enclaves/platforms and utilize built-in distributed communication and operational features (e.g., cross-cluster search (CCS)) to connect Elastic clusters and components. With regards to M-26-14 retention compliance, Elastic's available deployment options are as follows:

Fully hosted services:

- Elastic Cloud Hosted (ECH): Commercial AWS, Azure, and GCP
- ECH on GovCloud: Accredited to FedRAMP Moderate and High

Self-managed deployments:

- Elastic Cloud on Kubernetes (ECK): EKS, AKS, GKE, and standard Kubernetes distributions
- Self-hosted infrastructure: Physical, VM, and HCI hardware
- Air-gapped enclaves: Disconnected/isolated network systems

By virtue of its inherent distributed nature, Elastic allows organizations to deploy Elastic clusters and components near where their data is being generated, to index the data locally but then provide search-based access and analytics across remote clusters while still enforcing access controls on the data according to each cluster.

Data collection

Elastic can deploy collection components within existing infrastructure, to the edge, or to individual endpoints and send the collected data to the desired local, regional, or centralized Elasticsearch cluster. Elastic's primary data collection tools provided with the Elastic platform are:

- **Elastic Agent + Fleet + integrations:** Elastic Agent is a single, unified software agent deployed on host machines to collect logs, metrics, and security data, and to protect endpoints from threats. Fleet is the centralized management system in Kibana that allows you to configure, deploy, and monitor multiple Elastic Agents from a single location using policies. Agent policies can include any number of prebuilt integrations for collecting data from popular data providers, ad hoc endpoint inventory and metrics (via Osquery), as well as endpoint detection, protection, and remediation (EDR) capabilities Elastic Defend.
- **Beats + modules:** Lightweight purpose-built data shippers that reside on or near the edge and collect and send data downstream to be ingested into Elasticsearch. Beats modules are prepackaged configurations designed to simplify the collection, parsing, and visualization of data from common services and technologies. They automatically map raw data (like system logs or application metrics) into structured fields and provide prebuilt dashboards.
- **Logstash + input plugins:** Logstash is a data processing pipeline tool that ingests, transforms, and ships data. Input plugins are its entry points; they are modules that tell Logstash exactly where and how to collect data from various sources so it can be processed and sent to a destination.

While there are distinct advantages (such as automatic ECS conversion) to using the collection tools provided with the Elasticsearch Platform, data collection is not restricted to Elastic's tools. Other options include hybrid collection (where data is forwarded/centralized/queued and Elastic can use its tools to collect from that central collection), Elastic product OEM integrations, and index/connect-only methods (e.g., raw file system, ODBC/JDBC, or API-based collection).

Search, retrieval, and retention

Elastic is a search-based platform, with flexible ways to configure data retention and search responsiveness to suit the organization's mission requirements and cost constraints. Data is initially ingested into a fast data indexing and processing tier to perform normalizations, enrichments, and near real-time analytics. From there, depending on their use cases, many organizations will choose to either move the data into a slightly "colder" tier where data remains fully searchable and fast. But because the data is no longer being processed, only searched (and usually progressively less frequently over time), the main search tier can keep a longer searchable timeframe available. For use cases where data does not need to be near searchable to sub-second response times, Elastic offers a unique "frozen" (searchable snapshot) tier where the data is still fully searchable, but it's kept on object storage at might higher densities. For long-term retention, Elastic also offers standard snapshot and restore operations that will greatly reduce the time and effort required to "rehydrate" data that is being stored in a retention-only state for evidentiary or compliance purposes.

The Elastic M-26-14 reference architecture assumes that indexed data will flow automatically through index lifecycle management (ILM) policies and move through the Ingest (hot), (optional) warm Search, and Storage (frozen, yet fully searchable) tiers according to M-26-14 retention standards — at Optimal (Level 4) the requirement is ≥6 months of data remains searchable, and ≥12 months is retrievable. An added bonus of Elastic's fully searchable frozen Storage tier is that all data remains fully searchable for the entire ≥12 month retention period.

Additional required capabilities

In addition to the collection, retention, and retrieval/search on the logging data categories specified in Appendix B, the OMB memorandum included some additional capabilities considered necessary for full compliance with M-26-14 standards and maturity.

- For all Appendix B activity categories and Operational Technology (OT) and Internet of Things (IoT) devices/data sources:
    - Continuous Diagnostics and Mitigation (CDM), Hardware Asset Management (HWAM), and Software Asset Management (SWAM) inventory capabilities
    - Log collection, monitoring, analytics, and alerting
    - Identification and prioritization of High-Value Assets (HVA) and High-Impact Systems (HIS)
    - Sensitive data/PII protection
    - Identity/resource/data/event detection and alerting
    - NTP alignment and enforcement
    - Log integrity hashing/veracity
- Alignment to the CISA Zero Trust Maturity Model pillars
- Artificial Intelligence (AI) (and unsupervised machine learning) technologies for enhancing and automating analysis and detections
- Self-assessment of CEM and THIRF capabilities and maturity attestation

Elastic M-26-14 Concept of Operations

Progression through the M-26-14 Maturity Levels are primarily measured based on the following metrics and definitions:

- **Inventory visibility:** The percentage of the Agency's total IT/OT/IoT assets that have been captured in a centralized inventory (e.g., HWAM/SWAM)
- **Collection coverage:** The percentage of inventory that is actively collected, searchable, and/or retrievable according to the Agency's Logging Plan (due to CISA within 90 days after the LRA is published)
- **Collection operations:** The percentage of collected data that has active detection and alerting capabilities to support CEM and THIRF activities
- **Data retention (Searchable vs. Retrievable):** Each successive level of maturity requires a higher amount of (retrievable) data retention and searchability:
    - **Searchable:** Immediately available, actively searchable
    - **Retrievable:** Actively searchable after one or more intermediary steps
- **Log management:** Data storage and management standards increase with each maturity level, to include encryption (both at rest and in-transit), data integrity, JIT access, and gated deletion are required at Optimal (Level 4)

The combination of the Elasticsearch Platform's core features and the Elastic M-26-14 Compliance Pack lets organizations reach Advanced (Level 3) almost immediately and operate at Optimal (Level 4) within weeks. The Data Retention and Log Management elements (tiered ILM to a 12-month retrievable/6-month searchable window, encryption, integrity hashing, JIT access, two-gate retirement, and legal hold) are delivered on day one. Because overall maturity is set by the lowest-scoring element, the practical path to Optimal is driven by Inventory Visibility and Collection Coverage, which advance as the agency onboards assets and log sources.

The concept of operations is for Elastic to initially deploy its search-based platform with capacity to Optimal (Level 4) retention levels immediately, and progression through the Maturity Levels will proceed as data sources and detections are added to achieve compliance according to inventory, collection, and operational measures. For organizations that have existing functional systems in their ecosystem for data collection/retention, or for any of the additional required capabilities, the Elasticsearch Platform's integration strategy easily adapts to suit those existing investments.

Elastic M-26-14 Compliance Pack

The Elastic M-26-14 Compliance Pack includes assets of various types that are meant as a "quick-start" showcase for demonstrating how organizations can meet M-26-14 data collection, inventory, retention, detection, alerting, and self-assessment requirements with the Elasticsearch Platform.

Elastic M-26-14 Compliance Pack asset inventory

Category	Count	Key assets
Machine learning jobs	6	element1-asset-coverage, element2-ingestion-rate, element3-rule-silence, element4-ilm-anomaly, element5-hash-coverage, catb-dns-entropy
ML alert rules	13	E1–E5 element alerts + Cat-A/B/H supplementary
Detection rules	21 custom (+ ~1,800 prebuilt)	appendixb-a-* ... -k-* (12), ws7 posture-change r1–r7, retirement-gate1/2
Ingest pipelines	17	log-integrity-hash, alert-category-pipeline, osquery-normalize, asset-normalize / canonical-enrich / entity-store-enrich, baseline-hash
Transforms	12	score-entity, score-rollup / -enterprise, asset-entity-resolution, alert-coverage-daily / -latest, dataset-retention, store-feed, observed-assets
Enrich policies	8	authorized-software, asset-baseline-lookup, asset-compliance-lookup, inventory-enrich, config-enrich, host-alert-lookup, observed-enrich
ILM policies	9	logs-l3 / l4-hot-frozen, l3 / l4-no-delete, retention-l1 / l2 / l3, asset-inventory, logs-hva-extended

Compliance objective → Elastic delivery mechanism

Compliance objective	Elastic delivery mechanism
CEM - Log Collection	Elastic Agent + Fleet (with over 500+ integrations), Logstash, Beats

<table>
<tr><td>CEM - Retention Tiers</td><td>9 ILM policies: L3/L4 hot-frozen, L3/L4 no-delete, retention L1/L2/L3, asset inventory, HVA-extended</td></tr>
<tr><td>CEM - Asset Inventory</td><td>Direct CDM/HWAM/SWAM platform integrations and/or inventory via the Agent Osquery integration → HWAM/SWAM index template</td></tr>
<tr><td>CEM - Inventory & entity resolution</td><td>Osquery HWAM/SWAM/network pack, asset-entity-resolution + observed-assets transforms, 8 enrich policies</td></tr>
<tr><td>THIRF - Detection</td><td>21 custom M-26-14 detection rules + ~1,800 prebuilt Elastic Security rules</td></tr>
<tr><td>THIRF - Anomaly Detection</td><td>6 custom M-26-14 ML jobs + 13 ML alert rules</td></tr>
<tr><td>THIRF - Threat intelligence</td><td>CISA AIS + MS-ISAC TAXII 2.1 → Indicator Match rules</td></tr>
<tr><td>THIRF - Tamper-evident Logs</td><td>Log integrity hash ingest pipeline</td></tr>
<tr><td>THIRF - AI-assisted investigation & reporting</td><td>3 AI agents (threat-investigation, POA&M-drafting, after-action) + 4 ES|QL agent tools</td></tr>
<tr><td>Log Management (Optimal / L4)</td><td>Two-gate retirement (gate1/gate2 watchers + workflows), JIT access grants with jit-audit / jit-expiry, legal-hold selective-copy, encryption at rest/in-transit (ECH), SLM compliance snapshots |</td></tr>
<tr><td>Collection operations - Closed-loop actioning</td><td>Unknown-device-triage & data-classification-intake workflows</td></tr>
<tr><td>Appendix C Maturity Measurement</td><td>5 ML jobs (one per element) + Maturity Dashboard</td></tr>
<tr><td>Compliance Attestation Evidence</td><td>Compliance Attestation Dashboard + coverage matrix export</td></tr>
</table>

Architecture philosophy

Purpose-built node roles

The reference architectures in this document assign distinct Elasticsearch node roles to distinct hardware pools. Rather than running all functions on generalist nodes, each role — hot, cold, frozen, master, and ML — runs on instance types optimized for its specific workload. This separation eliminates resource contention, enables independent scaling of each tier, and provides clear cost attribution.

The data lifecycle: Hot → cold → frozen

Data moves through storage tiers as it ages, trading query speed for cost efficiency. Fresh data lives on high-performance hot nodes (e.g., NVMe SSD-backed c7gd instances) where it is actively indexed and queried. As data ages past its hot retention window, it transitions to cold nodes (e.g., i3en), which offer high-density spinning storage at significantly lower cost per TB. Finally, data that must be retained for compliance or historical analysis but is rarely queried moves to frozen tier, backed by object storage. Frozen data is retrieved on demand using searchable snapshots, which mount indices directly from object storage without requiring a full restore — enabling near-unlimited retention at minimal cost. For data that must satisfy the retrievable requirement under M-26-14 (minimum 12 months), unmounted snapshots provide a complementary approach: indices remain in object storage in an unattached state and are not actively searchable but can be fully mounted and made queryable on demand within the timeframes the directive requires. This distinction allows organizations to architect a cost-appropriate response to each retention tier — frozen searchable snapshots for CEM-aligned data that may be queried at any time, and unmounted snapshots for THIRF-aligned data that simply needs to be recoverable when called upon.

This tiered approach is what allows the architecture to scale economically with retention requirements. A cluster retaining 12 months of data does not require 12 months of hot storage — only the most recent window needs to be immediately accessible. The result is a cost profile that grows sub-linearly with retention period.

Master nodes: Cluster stability above all else

Dedicated master nodes are responsible solely for cluster coordination: tracking shard allocation, managing cluster state, and handling node join/leave events. They are sized conservatively and never serve data. Three master nodes are deployed across three AZs to maintain quorum and prevent split-brain scenarios. This count remains constant across all t-shirt sizes — master node requirements are driven by cluster complexity, not data volume.

Machine learning nodes: Isolated inference capacity

Machine learning nodes run anomaly detection, inference pipelines, and data frame analytics workloads without competing with search and ingest traffic. By isolating ML on dedicated m5dn instances, customers can scale analytical capacity independently and avoid the latency spikes that occur when ML jobs share resources with query-serving nodes. ML node RAM scales with cluster size, increasing from 15 GB for small deployments to 60 GB for large ones, reflecting the greater ML workload that typically accompanies higher-volume environments.

Multi-availability zone (AZ) deployment

All cluster sizes deploy across three availability zones. Shards are distributed across AZs so that the loss of any single zone does not result in data loss or significant degradation of query performance. This applies to all node tiers — hot, cold, frozen, master, and ML nodes each maintain AZ-aware placement.

Node role reference

The following table summarizes the purpose, instance type example, and behavior of each node role used in this architecture.

Node role	Instance (example)	Description
Hot	c7gd	Handles all active indexing and serves recent, high-frequency queries. NVMe SSD-backed for maximum I/O throughput. This tier holds the most recently ingested data within the configured hot retention window. Node count scales directly with ingest rate.
Cold	i3en	Stores older indices that are queried infrequently. High-capacity HDD-backed instances provide a significant cost reduction per TB compared to hot nodes. Data here is fully searchable but with higher latency than hot. Indices are moved here when they age out of the hot retention window.
Frozen	i3en	Provides low-cost, long-term retention using searchable snapshots stored in S3. Frozen data is mounted on demand and is not kept in local disk cache between queries. Ideal for compliance-driven retention where data may need to be retrieved occasionally but not continuously. Effectively decouples storage cost from retention period.
Master	c7gd	Manages cluster-wide state: shard allocation, node membership, and index metadata. Always deployed as a set of three dedicated nodes across three AZs. Never assigned data or ML roles. Constant across all t-shirt sizes at 15 GB RAM.
ML	m5dn	Runs anomaly detection jobs, inference pipelines, and data frame analytics. Isolated from search and ingest traffic. RAM scales with cluster size (15–60 GB) to accommodate the increasing ML workloads that accompany larger deployments.
Kibana	c7gd	Serves the Kibana UI and handles visualization, dashboard rendering, and user session management. Allocated 30 GB RAM across all cluster sizes. Kibana workload is largely UI-driven and does not scale linearly with ingest volume.

Reference architecture definitions

Each cluster size is defined by its maximum sustained ingest rate and the corresponding storage footprint. Customers should select the size that covers their peak ingest rate with headroom — right-sizing to the median can result in degraded performance during spikes. All configurations below are for ECH on AWS within a FedRAMP High boundary.

SMALL

• *Up to 2 TB/day ingest* • Up to 403 TB object storage

Typical for departmental SIEM, dev/test, or early-stage observability platforms

Profile

The Small cluster is designed for teams standing up their first production Elasticsearch environment, or for workloads with well-defined, moderate-scale data requirements. At up to 2 TB/day ingest, this tier handles a typical departmental security operations center (SOC), a mid-size application observability stack, or a dev/test environment requiring production-grade architecture. The Small configuration provides full multi-tier ILM and multi-AZ resilience without the operational overhead of a larger footprint.

Node configuration

Node role	Instance (example)	Node count	RAM	Purpose
Hot	c7gd	3–6	60 GB	Active indexing and recent query serving; 3 nodes at 0.5–1 TB/day, 6 nodes at 2 TB/day
Cold	i3en	3	60 GB	Older indices with reduced query frequency; minimum of 3 for AZ distribution
Frozen	i3en	3–6	60 GB	Long-term S3-backed retention; grows to 6 nodes at 2 TB/day
Master	c7gd	3	15 GB	Dedicated cluster coordination; constant across all sizes
ML	m5dn	0–1	15 GB	Anomaly detection and inference; 15 GB accommodates standard ML jobs at this scale
Kibana	c7gd	1	30 GB	UI and dashboard serving; constant across all sizes

ILM Policy Guidance

Tier	Rollover/move trigger	Recommended retention	Storage recommendation
Hot	50 GB shards or 1 day	1 day	NVMe SSD
Cold	Move from hot after 1 day	7 days	HDD
Frozen	Move from cold after 7 days	357 days	Object storage

Constraints and considerations

- Not recommended for ingest rates that regularly exceed 2 TB/day sustained — burst above this threshold may saturate hot nodes
- ML node RAM of 15 GB limits concurrent anomaly detection jobs; complex or large-model inference should be evaluated against available capacity
- Suitable for a single team or business unit; multi-team shared environments with high concurrent query loads should consider Medium

MEDIUM

• *2–10 TB/day ingest* • Up to ~2,015 TB object storage

Typical for enterprise SOC, multiteam observability, or regulated data platforms

Profile

The Medium cluster serves enterprise environments where multiple teams or workloads share a common Elasticsearch platform. At up to 10 TB/day ingest, this tier is well-suited for an enterprise security operations center ingesting logs across a large infrastructure estate, a multi-application observability platform, or a compliance-driven data platform with extended retention requirements. Hot node counts scale from 6 to 27 across the medium range, with cold and frozen tiers growing proportionally.

Node Configuration

Node role	Instance (example)	Node count	RAM	Purpose
Hot	c7gd	6–27	60 GB	Scales with ingest: 6 nodes at 2 TB/day through 27 nodes at 10 TB/day
Cold	i3en	3–12	60 GB	Grows from minimum 3-node base to 12 nodes as retained data volume increases
Frozen	i3en	6–21	60 GB	Expands to accommodate multi-month object-backed retention at enterprise scale
Master	c7gd	3	15 GB	Unchanged from Small; cluster coordination requirements are not data-volume dependent
ML	m5dn	0–1	15–30 GB	Scales to 30 GB at higher ingest rates to support broader anomaly detection coverage
Kibana	c7gd	1	30 GB	UI and dashboard serving; constant across all sizes

ILM policy guidance

Tier	Rollover/move trigger	Recommended retention	Storage recommendation
Hot	50 GB shards or 2 days	2 days	NVMe SSD
Cold	Move from hot after 2 days	7 days	HDD
Frozen	Move from cold after 7 days	356 days	Object storage

Constraints and considerations

- At the upper end of the medium range (8–10 TB/day), customers should plan for a Large upgrade within 6–12 months if ingest continues — or is planned — to grow.
- ML RAM scales to 30 GB at higher ingest rates; customers running extensive ML pipelines (e.g., LLM inference, large model scoring) should validate against available capacity before sizing.
- Cold and frozen node counts should be validated against actual retention policy — the figures above assume standard 7-day cold + 12 months frozen retention. Longer hot retention will require additional cold nodes.

LARGE

• *10–25 TB/day ingest* • Up to ~5,037 TB object storage

Typical for large federal agencies, enterprise-wide SIEM, or high-volume data platforms

Profile

The Large cluster is designed for high-volume, mission-critical deployments where Elasticsearch serves as the backbone of an enterprise-wide or agency-wide data platform. At up to 25 TB/day ingest, this tier accommodates large federal agency security programs, high-throughput observability platforms spanning thousands of endpoints and services, or any environment where data volumes are high and retention requirements are at the upper bound of what is required by M-26-14 Maturity Level 4. Hot node counts reach 66 at peak, with cold and frozen tiers scaling to 30 and 54 nodes respectively.

Node Configuration

Node role	Instance (example)	Node count	RAM	Purpose
Hot	c7gd	27–66	60 GB	27 nodes at 10 TB/day scaling to 66 at 25 TB/day; NVMe SSD for maximum write throughput
Cold	i3en	12–30	60 GB	High-density HDD nodes storing several days of aged indices
Frozen	i3en	21–54	60 GB	Large object-snapshot fleet for petabyte-scale long-term retention
Master	c7gd	3	15 GB	Constant at 3 nodes; cluster complexity at this scale still does not require more than 3 dedicated masters
ML	m5dn	0–1	15–60 GB	Scales to 60 GB at higher ingest rates to support broader anomaly detection coverage
Kibana	c7gd	1	30 GB	UI and dashboard serving; constant across all sizes

ILM policy guidance

Tier	Rollover/move trigger	Recommended retention	Storage recommendation
Hot	50 GB shards or 3 days	3 days	NVMe SSD
Cold	Move from hot after 3 days	7 days	HDD
Frozen	Move from cold after 7 days	355 days	Object storage

Constraints and considerations

- Ingest rates above 25 TB/day sustained will require a custom sizing engagement — the Large tier represents the upper bound of this standard reference model.
- At this scale, index lifecycle management configuration has a significant cost impact. Hot retention windows should be minimized to control c7gd node count.
- ML at 60 GB RAM supports extensive workloads; customers planning to run large language model inference or real-time scoring at scale should discuss dedicated ML fleet sizing with their solutions architect.
- Capacity planning reviews are recommended quarterly at the Large tier due to the cost sensitivity of incremental node additions at this scale.

At-a-glance reference architecture comparison

The table below summarizes all three reference architectures for quick reference and side-by-side comparison.

	Small	Medium	Large
Ingest rate	0.5–2 TB/day	2–10 TB/day	10–25 TB/day
S3 data stored	Up to ~403 TB	Up to ~2,015 TB	Up to ~5,037 TB
Hot nodes (c7gd)	3–6	6–27	27–66
Cold nodes (i3en)	3	3–12	12–30
Frozen nodes (i3en)	3–6	6–21	21–54
ML node RAM (m5dn)	15 GB	15–30 GB	30–60 GB
Master node RAM (c7gd)	15 GB (×3)	15 GB (×3)	15 GB (×3)
Kibana RAM (c7gd)	30 GB	30 GB	30 GB
Availability zones	3	3	3

Master node count is always 3 across all sizes. RAM of 15 GB per master node is allocated on c7gd instances. Kibana is always allocated 30 GB RAM on c7gd. Node counts for hot, cold, and frozen represent the range across the ingest rates that fall within each tier.

Sizing decision guide

Use the following questions to identify the appropriate starting tier. When in doubt, size up — it is easier to right-size down after initial deployment than to remediate an undersized cluster under load.

Step 1: What is your peak sustained ingest rate (TB/day)?

Ingest rate	Resulting architecture
≤ 2 TB/day	Small
2–10 TB/day	Medium
10–25 TB/day	Large
> 25 TB/day	Contact Elastic for custom sizing

Step 2: What are your M-26-14 retention obligations?

Maturity Level	Retention requirement	Sizing guidance
Level 1 — Initial	Retrievable minimum 6 months	A Small cluster with frozen tier configured for unmounted snapshots is likely sufficient
Level 2 — Intermediate	Retrievable minimum 12 months	Small may still be appropriate depending on ingest rate; validate frozen node counts against expected S3-stored volume before committing
Level 3 — Advanced	Searchable minimum 3 months + retrievable 12 months	Confirm hot and cold tier sizing supports a 3-month searchable window; customers near the upper bound of Small should consider Medium
Level 4 — Optimal	Searchable minimum 6 months + retrievable 12 months	Validate that hot and cold tiers combined cover the full 6-month searchable window; customers near the upper bound of their current tier should size up to ensure headroom
Beyond M-26-14	Agency records schedule or mission-driven retention beyond 12 months	Frozen tier with extended S3 retention is essential regardless of cluster size; treat extended retention requirements as a strong signal to move to Medium or Large if not already there

Step 3: What are your CEM and THIRF workload characteristics?

Workload profile	Characteristics	Sizing guidance
CEM-primary	Real-time SOC monitoring, alerting, near-term investigation	Hot and cold tier sizing is the primary cost and performance driver; if ingest rate falls within the Small range but query concurrency is high, consider starting at Medium to preserve headroom
THIRF-heavy	Frequent forensic lookups, threat hunting across historical data, incident reconstruction	Frozen node counts must support concurrent retrieval load, not just storage volume; customers with high retrieval demand may require Medium or Large even at lower ingest rates
ML-intensive	Anomaly detection, inference pipelines, AI-enhanced detection per M-26-14 Level 4 guidance	Standard anomaly detection is well-served by default ML allocations at any tier; extensive pipelines or large-model inference are a common reason to select Medium over Small independent of ingest rate

Growth and upgrade considerations

The reference architectures in this document are starting points, not hard ceilings. Elasticsearch scales horizontally and upgrades between tiers are non-disruptive when planned appropriately. The following signals indicate it is time to revisit your cluster size.

Signals that you are approaching the limits of your current tier:

- Hot node CPU utilization consistently above 70% during peak ingest windows

- Indexing latency increasing without a corresponding spike in ingest volume
- JVM heap pressure on data nodes exceeding 75% sustained
- Query latency degrading during ingest spikes (indexing and query competing for resources)
- S3 stored data approaching the upper bound of your tier's reference range

What an upgrade looks like

Moving from Small to Medium, or Medium to Large, involves adding nodes to each tier. Because ECH manages shard rebalancing automatically, adding hot, cold, or frozen nodes is an online operation — existing shards are redistributed to new nodes without downtime. Master nodes remain at 3 regardless of tier. The upgrade process involves updating the deployment configuration in the ECH console, and Elastic handles the orchestration. Customers should plan capacity reviews quarterly and proactively size up before hitting a ceiling. Reactive scaling under load is possible but carries operational risk during the transition period.

